A Compact Ordered Metric Space Which Is Not L-Ordered

Run `fa_banach_001`, lane 17

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Source Problem

The source paper is T. Fritz and P. Perrone, *Stochastic order on metric spaces and the ordered Kantorovich monad*, arXiv:1808.09898. In Section 4.1, after introducing L-ordered spaces, the authors ask:

> Problem 4.1.3. Is every compact ordered metric space automatically L-ordered?

The problem appears on page 18 of the arXiv PDF.

> Many ordered metric spaces of interest in mathematics are L-ordered. In particular, as we will show, L-ordered metric spaces are precisely the spaces which are embeddable into an ordered Banach space (see Remark 6.1.5). So far we have not yet been able to find any non-L-ordered space which is also compact:
>
> **Problem 4.1.3.** *Is every compact ordered metric space automatically L-ordered?*
>
> In the following, we denote by L-OMet and L-COMet the full subcategories of OMet

Definitions

Following the source paper, an ordered metric space is a metric space equipped with a partial order whose graph is closed in the product metric topology. It is L-ordered if, for every $x, y \in X$,

$$x \leq y \quad \Longleftrightarrow \quad f(x) \leq f(y) \text{ for every short monotone } f : X \to \mathbb{R}.$$

Here short means 1-Lipschitz.

Proof Intuition

Compactness prevents the source paper's noncompact counterexample from being copied directly: if a short metric bridge collapses to a genuine limit point, closedness of the order tends to create new order relations. The example below avoids this by placing many finite "almost chains" inside the compact interval. Each link is an actual order relation followed by a tiny metric gap; the total metric error in the nth chain tends to zero, so every short monotone real-valued function is forced to satisfy $f(0) \leq f(1)$.

At the same time, the order relation itself contains no composable nontrivial pairs. All nontrivial lower endpoints are rational and all nontrivial upper endpoints are irrational. Thus the order is transitive for a simple discrete reason, while closedness holds because the off-diagonal pairs approach only the diagonal.

Theorem 1. *There exists a compact ordered metric space which is not L-ordered. In fact, the usual compact metric space* $[0,1]$ *admits such a closed partial order.*

Proof. Let $X = [0,1]$ with its usual metric. For every integer $n \geq 2$, put

$$\varepsilon_n = \frac{\sqrt{2}}{n^3}.$$

For $0 \leq i \leq n-1$, define

$$a_{n,i} = \frac{i}{n}, \qquad b_{n,i} = \frac{i+1}{n} - \varepsilon_n.$$

Since $0 < \varepsilon_n < 1/n$, all $b_{n,i}$ lie in $[0,1]$. Each $a_{n,i}$ is rational, while each $b_{n,i}$ is irrational.

Define a relation $\leq$ on X by declaring

$$x \leq y \quad \iff \quad x = y \text{ or } (x,y) = (a_{n,i}, b_{n,i}) \text{ for some } n \geq 2,\ 0 \leq i \leq n-1.$$

This relation is reflexive by construction. It is antisymmetric because a nontrivial related pair has rational first coordinate and irrational second coordinate, so the reverse pair cannot also be nontrivial and equality is impossible.

It is transitive as well. Suppose $x \leq y$ and $y \leq z$. If either relation is equality, the conclusion is immediate. Otherwise, $x \leq y$ is a nontrivial pair, so y is irrational; but every first coordinate of a nontrivial pair is rational. Hence $y \leq z$ cannot be nontrivial, a contradiction to the remaining case. Therefore $\leq$ is a partial order.

We next prove that its graph is closed. Let

$$E = \{(a_{n,i}, b_{n,i}) : n \geq 2,\ 0 \leq i \leq n-1\}.$$

The graph is $\Delta_X \cup E$, where Δ_X is the diagonal. Consider a convergent sequence in E, say

$$(a_{n_k,i_k}, b_{n_k,i_k}) \to (u,v).$$

If the integers n_k are bounded, then only finitely many points of E can occur, and a constant subsequence shows that $(u,v) \in E$. If $n_k \to \infty$, then

$$|b_{n_k,i_k} - a_{n_k,i_k}| = \frac{1}{n_k} - \varepsilon_{n_k} \longrightarrow 0,$$

so $u = v$ and $(u,v) \in \Delta_X$. Thus $\Delta_X \cup E$ is closed in $X \times X$. Hence X is a compact ordered metric space.

We now show that X is not L-ordered. Let $f : X \to \mathbb{R}$ be any short monotone map. Fix $n \geq 2$. Monotonicity gives

$$f(a_{n,i}) \leq f(b_{n,i})$$

for all $0 \leq i \leq n-1$. Since f is short and $|b_{n,i} - a_{n,i+1}| = \varepsilon_n$, we also have

$$f(b_{n,i}) \leq f(a_{n,i+1}) + \varepsilon_n.$$

Combining these inequalities yields

$$f(a_{n,i}) \leq f(a_{n,i+1}) + \varepsilon_n.$$

Iterating from $i = 0$ to $i = n - 1$ gives

$$f(0) = f(a_{n,0}) \le f(a_{n,n}) + n\varepsilon_n = f(1) + \frac{\sqrt{2}}{n^2}.$$

Letting $n \to \infty$ gives $f(0) \le f(1)$ for every short monotone $f : X \to \mathbb{R}$.

However, $0 \not\le 1$ in the order: the pair $(0, 1)$ is neither diagonal nor one of the pairs $(a_{n,i}, b_{n,i})$, since all such second coordinates are strictly less than 1 when $i = n - 1$ and are even smaller otherwise. Thus short monotone functions fail to detect the order, and X is not L-ordered. $\square$

Verification Notes

The proof is symbolic and uses no computational search. The key verifications are:

- $\varepsilon_n = \sqrt{2}/n^3$ is irrational and satisfies $0 < \varepsilon_n < 1/n$ for $n \ge 2$.
- Nontrivial first coordinates are rational, while nontrivial second coordinates are irrational, so no two nontrivial order pairs compose.
- The only possible new accumulation points of nontrivial order pairs are diagonal points, because their coordinate gaps are $1/n - \varepsilon_n \to 0$.
- The finite-chain estimate $f(0) \le f(1) + n\varepsilon_n = f(1) + \sqrt{2}/n^2$ holds for every short monotone f.

No simple computational contradiction is possible here: the construction is explicit, compactness is inherited from the usual interval, and all order checks are contained in the preceding proof.

Novelty Check and Limitations

The local run indexes were searched for arXiv:1808.09898, the source title, and phrases including `compact ordered metric space`, `L-ordered`, `short monotone`, and `ordered Kantorovich`. No existing solution packet or proof-gap entry for this question was found. The local arXiv source corpus was searched for the same phrases; it found the source paper and related terminology but no later answer to Problem 4.1.3.

External web searches were also run for the exact sentence "Is every compact ordered metric space automatically L-ordered" and close variants involving L-ordered metric spaces, compact ordered metric spaces, counterexamples, and short monotone maps. These searches returned the source paper but no explicit later solution. This is a bounded novelty check, not a publication-level exhaustive literature review.

The counterexample answers the exact problem as stated in the source paper. It uses a highly nonstandard closed order on the standard compact metric interval; it does not contradict the source paper's positive classes such as subsets of ordered Banach spaces.

Human-Review Recommendation

This should be reviewed as a likely valid counterexample. The reviewer should focus on the closedness and transitivity of the sparse relation, and on the iteration of the Lipschitz bridges from 0 to 1.

References

[1] T. Fritz and P. Perrone, *Stochastic order on metric spaces and the ordered Kantorovich monad*, arXiv:1808.09898.